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Sensitivity analysis of energy inputs for barley production in Hamedan Province of Iran

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ABSTRACT

The objectives of this study were to determine the energy consumption and evaluation of inputs sensitivity for barley production in Hamedan Province, Iran. The sensitivity of energy inputs was estimated using the marginal physical productivity (MPP) method and partial regression coefficients on barley yield. The results revealed that total energy input for barley production was ~25,027 MJ ha^{-1}; the non-renewable energy shared about 66% while the renewable energy did 34%. Energy use efficiency, energy productivity, and net energy were 2.86, 0.19 kg MJ^{-1}, and ~46,498 MJ ha^{-1}, respectively. Econometric model evaluation showed that machinery energy was the most significant input which affects the output level. Sensitivity analysis indicates that with an additional use of 1 MJ of each of the human labour, machinery and electricity energy would lead to an increase in yield by 7.37, 1.66 and 0.33 kg, respectively. Also, the MPP of biocides energy was calculated to be −1.97 implying that the use of biocides energy is in excess for barley production, causing an environmental risk problem in the region.

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1. Introduction

Barley is a major staple food in several regions of the world and food barley is generally found in regions where other cereals do not grow well due to altitude, low rainfall, or soil salinity. It remains the most viable option in dry areas (<300 mm of rainfall). Food barley is used either for bread making (usually mixed with bread wheat) or for specific recipes (FAO, 2007). Barley production in Iran was 3 million tonnes in 2007 (FAO, 2007), from which 7% is produced in Hamedan Province (Anonymous, 2007). Wheat, alfalfa, barley, potato, and corn are dominant products grown in this province.

Efficient use of energy is one of the principal requirements of sustainable agriculture. Energy use in agriculture has been increasing in response to increasing population, limited supply of arable lands, and a desire for higher standards of living. Continuous demand in increasing food production resulted in intensive use of energy inputs and natural resources. However, intensive use of energy causes problems threatening public health and environment. Efficient use of energy in agriculture will minimize environmental problems, prevent destruction of natural resources, and promote sustainable agriculture as an economical production system (Erdal et al., 2007). Economic production is a function of many factors such as human labour, capital, natural resources, availability of energy and technology. Therefore, both the natural resources are rapidly decreasing and the amount of contaminants is considerably increasing. The best way to lower the environmental hazard of energy use is to increase the energy use efficiency (Esengun et al., 2007). Energy input–output analysis is usually used to evaluate the efficiency and environmental impacts of production systems. It is also used to compare the different production systems.

Energy consumption by the agriculture sectors can be broadly categorized into direct and indirect energy use. Agriculture uses energy directly as fuel or electricity to operate machinery and equipment, to heat or cool buildings and for lighting on the farm and indirectly in the chemical fertilizers, seed production, machinery and biocides produced off the farm (Uhlin, 1998; Ozkan et al., 2004). Considerable studies have been conducted on energy use in agriculture (Kuesters and Lammel, 1999; Jianbo, 2006; Strapatsa et al., 2006; Uzunoz et al., 2008; Kizilaslan, 2009), while there are few studies on the energy requirements and sensitivity analysis of inputs in barley production.

In a research conducted in Spain, energy use and economic evaluation were considered for winter wheat, winter barley, spring barley and vetch production. The spring barley showed highest energy consumption since a larger number of tillage operations were required and a larger amount of herbicides for weed control (Hernanz et al., 1995). Also, Khan et al. (2009) studied the energy inputs in wheat, rice and barley production for reducing the environmental footprint of food production in Australia. The results showed that barley crop seems more efficient in terms of energy and water use jointly.

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Nomenclature

n	required sample size
N	number of holdings in target population
s	standard deviation
D	acceptable error (permissible error was chosen as 5%)
T	confidence limit (1.96 in the case of 95% reliability)
Y_i	yield level of the ith farmer
X_1	labour energy
X_2	machinery energy
X_3	diesel fuel energy
X_4	total fertilizer energy
X_5	biocides energy
X_6	water for irrigation energy
X_7	electricity energy
X_8	seed energy
e_i	error term
α_i	coefficients of the exogenous variables
β_i	coefficients of the exogenous variables
γ_i	coefficients of the exogenous variables
DE	direct energy
IDE	indirect energy
RE	renewable energy
NRE	non-renewable energy
MPP_{xj}	marginal physical productivity of jth input
α_j	regression coefficient of jth input
$GM(Y)$	geometric mean of yield
$GM(X_j)$	geometric mean of jth input energy

This study was aimed to determine input–output energy use in barley production in Hamedan, Iran from efficiency of energy consumption point of view. Also, the Cobb–Douglas production function was used to study the sensitivity and the relationship between energy inputs and barley yield.

2. Materials and methods

The research was done in Hamedan Province which is located in the west of Iran; within 33°59′ and 35°48′ north latitude and 47°34′ and 49°36′ east longitude. Hamedan region has an area of 1,949,400 ha; and the farming area, with a share of 51.7%, is 1,008,038 ha (Anonymous, 2007). The data used in the study were collected from 67 barley farms in 20 villages using a face-to-face questionnaire in March and April 2009. The collected data belonged to the production period of 2007–2008.

The simple random sampling method was used to determine the survey volume, as described by Kizilaslan (2009):

$$n = \frac{N(s \times t)^2}{(N-1)d^2 + (s \times t)^2} \tag{1}$$

Thus, calculated sample size in this study was found to be 67.

The inputs used in the production of barley were specified in order to calculate the energy equivalences in the study. Inputs in barley production were: human labour, machinery, diesel fuel, chemical fertilizers, farmyard manure, pesticides, fungicides, herbicides as biocides, water for irrigation, and electricity. The output was considered barley seeds. The energy equivalents given in Table 1, were used to calculate the input amounts.

The input and output were calculated per hectare and then, these input and output data were multiplied by the coefficient of energy equivalent. Following the calculation of energy input and output values, the energy ratio (energy use efficiency), energy productivity and net energy were determined (Mandal et al., 2002; Mohammadi et al., 2008):

$$\text{Energy use efficiency} = \frac{\text{Energy output (MJ ha}^{-1})}{\text{Energy input (MJ ha}^{-1})} \tag{2}$$

$$\text{Energy productivity} = \frac{\text{Barley output (kg ha}^{-1})}{\text{Energy input (MJ ha}^{-1})} \tag{3}$$

$$\text{Net energy} = \text{Energy output (MJ ha}^{-1}) - \text{energy input (MJ ha}^{-1}) \tag{4}$$

The energy use efficiency is the ratio between the output products and the total sequestered energy in the production inputs. The energy use efficiency gives an indication of how much energy was produced per unit of energy utilized. The energy productivity provides quantitative data on how much barley is obtained per unit of input energy. Net energy is defined as the difference between the gross energy output produced and the total energy used for obtaining it.

The Cobb–Douglas production function yielded the best estimates in terms of statistical significance and expected signs of parameters. The Cobb–Douglas production function is expressed as:

$$Y = f(x)\ \exp(u) \tag{5}$$

This function has been used by several authors to examine the relation between energy inputs and yield (Singh et al., 2004; Hatrili et al., 2006; Mohammadi and Omid, 2010). Eq. (5) can be linearized

Table 1
Energy equivalent of inputs and output in agricultural production.

Inputs (unit)	Unit	Energy equivalent (MJ unit^{-1})	Reference
A. Inputs			
1. Human labour	h	1.96	Yilmaz et al. (2005)
2. Machinery	h	62.70	Singh (2002)
3. Diesel fuel	l	56.31	Singh (2002)
4. Total fertilizer			
(a) Nitrogen	kg	66.14	Shrestha (1998)
(b) Phosphate (P_2O_5)	kg	12.44	Shrestha (1998)
(c) Farmyard manure	kg	0.30	Singh (2002)
5. Biocides	kg	120	Singh (2002)
6. Water for irrigation	m^3	1.02	Acaroglu (1998)
7. Electricity	kWh	11.93	Ozkan et al. (2004)
8. Seed	kg	14.7	Ozkan et al. (2004)
B. Output			
1. Barley	kg	14.7	Ozkan et al. (2004)

and re-written as:

$$\ln\ Y_i = a + \sum_{j=1}^{n} \alpha_j \ln(X_{ij}) + e_i, \quad i = 1, 2, \ldots, n \tag{6}$$

Eq. (6) can be expressed in the following form:

$$\ln\ Y_i = a_0 + \alpha_1 \ln X_1 + \alpha_2 \ln X_2 + \alpha_3 \ln X_3 + \alpha_4 \ln X_4 + \alpha_5 \ln X_5 + \alpha_6 \ln X_6 + \alpha_7 \ln X_7 + \alpha_8 \ln X_8 + e_i \tag{7}$$

Using Eq. (7), the effect of energy inputs on barley yield for each input was studied. On the other hand, barley yield (endogenous variable) was assumed to be a function of human labour, machinery, diesel fuel, total fertilizer, biocides, water for irrigation, electricity and seed energy (exogenous variables).

The effect of direct, indirect, renewable and non-renewable energies on production was also studied. For this purpose, Cobb–Douglas function was determined as (8) and (9):

$$\ln Y_i = \beta_0 + \beta_1 \ln DE + \beta_2 \ln IDE + e_i \tag{8}$$

$$\ln Y_i = \gamma_0 + \gamma_1 \ln RE + \gamma_2 \ln NRE + e_i \tag{9}$$

Eqs. (7)–(9) were estimated using ordinary least square technique.

The MPP method, based on the response coefficients of the inputs was utilized to analyze the sensitivity of energy inputs on barley yield. Sensitivity analysis is especially useful in pinpointing which assumptions are appropriate variables for additional data collection to narrow the degree of uncertainty in the results. Typically, in a sensitivity analysis, as the exogenous parameters are generally varied by a linear proportion, the endogenous variable must linearly depend on those parameters. Also, as the parameters are varied one at a time, different model parameters must not interact in their influence on the endogenous variable (Drechsler, 1998). Therefore, the sensitivity analysis of an input imposes the change in the output level with a unit change in the input in model, assuming that all other inputs are constant at their geometric mean level. The MPP of the various inputs was computed using the α_j of the various energy inputs as (Singh et al., 2004):

$$MPP_{xj} = \frac{GM(Y)}{GM(X_j)} \times \alpha_j \tag{10}$$

In production, returns to scale refer to changes in output subsequent to a proportional change in all inputs (where all inputs increase by a constant factor). In the Cobb–Douglas production function, it is indicated by the sum of the elasticities derived in the form of regression coefficients. If the sum of the estimated coefficients is greater than unity ($\sum_{i=1}^{n} \alpha_i > 1$), then it could be concluded that the increasing returns to scale, on the other hand if the latter parameter is less than unity ($\sum_{i=1}^{n} \alpha_i < 1$), then it is indicated that the decreasing returns to scale; and, if the result is unity ($\sum_{i=1}^{n} \alpha_i = 1$), it shows that the constant returns to scale (Singh et al., 2004).

Basic information on energy inputs and barley yields were entered into Excel's spreadsheet and simulated using Shazam 9.0 software.

3. Results and discussion

3.1. Analysis of input–output energy use in barley production

The inputs used in barley production and their energy equivalents with output energy rates are shown in the Table 2. The results revealed that around 83 h of human labour and 18 h of machinery power per hectare were required to produce barley in the research area. The amount of total fertilizers and biocides used for barley growing was ~1491 and 1.53 kg ha^{-1}, respectively. The total energy input for various processes in the barley production was calculated to be ~25,027 MJ ha^{-1} (Table 2). Nguyen and Haynes (1995) concluded that the input energy for barley production in Conventional and organic farming were to be 82,384 and 71,020 MJ ha^{-1}, respectively.

Table 2
Amounts of inputs, output and energy inputs and output for barley production in Hamedan, Iran.

Inputs (unit)	Quantity per unit area (ha)	Total energy equivalent (MJ ha^{-1})
A. Inputs		
1. Human labour (h)	83.27	163.21
2. Machinery (h)	18.22	1142.39
3. Diesel fuel (l)	104.13	5863.56
4. Total fertilizer (kg)	1490.97	6935.36
(a) Nitrogen (kg)	84.43	5584.20
(b) Phosphate (P_2O_5) (kg)	76.54	952.16
(c) Farmyard manure (kg)	1330.00	399.00
5. Biocides (kg)	1.53	183.60
6. Water for irrigation (m^3)	4674.52	4768.01
7. Electricity (kWh)	232.21	2770.27
8. Seeds (kg)	217.76	3201.07
The total energy input (MJ)		25027.47
B. Output		
1. Barley (kg)	4865.67	71525.37
Total energy output (MJ)		71525.37

The average inputs energy consumption was highest for total fertilizer (~6935 MJ ha^{-1}) including chemical fertilizers (~6536 MJ ha^{-1}) and farmyard manure (399 MJ ha^{-1}) which accounted for about 27% (Fig. 1) of the total energy input, followed by diesel fuel (~5864 MJ ha^{-1}, 23%). Fig. 1 shows the percentage distribution of the energy associated with the inputs. It has also been reported that the energy input of chemical fertilizers has the biggest share of the total energy input in agricultural crops production (Tsatsarelis, 1993; Erdal et al., 2007; Uzunoz et al., 2008; Kizilaslan, 2009). The shares of nitrogen and phosphorus energy were around 85% and 15%, respectively, from the total energy of chemical fertilizer used. The inputs energy consumption was least for human labour (~163 MJ ha^{-1}) which accounted for about 0.65% of the total energy consumption (Fig. 1). Similar results have been reported in several studies (Sartori et al., 2005; Kizilaslan, 2009; Strapatsa et al., 2006). The average yield of barley was obtained to be ~4866 kg ha^{-1}, accordingly, the total energy output per hectare were calculated ~71,525 MJ ha^{-1} (Table 2).

The energy use efficiency, energy productivity and net energy gain of barley production in the Hamedan Province are listed in Table 3. The energy use efficiency in the production of barley was

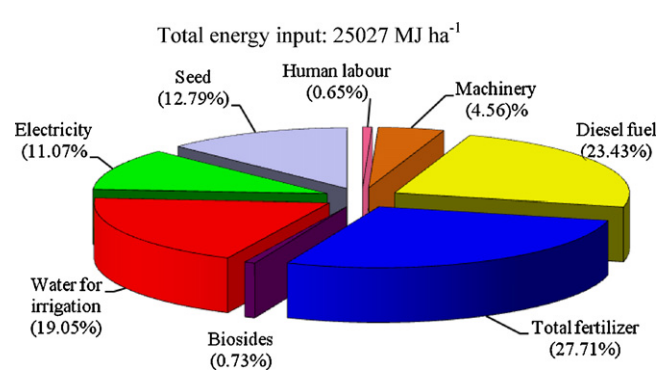

Fig. 1. The share of energy inputs for barley production in Hamedan, Iran.

Table 3
Some energy parameters in barley production.

Items	Unit	Barely
Energy use efficiency	–	2.86
Energy productivity	kg MJ^{-1}	0.19
Net energy gain	MJ ha^{-1}	46497.90
Direct energy[a]	MJ ha^{-1}	13565.05 (54.20%)
Indirect energy[b]	MJ ha^{-1}	11462.42 (45.80%)
Renewable energy[c]	MJ ha^{-1}	8531.29 (34.09%)
Non-renewable energy[d]	MJ ha^{-1}	16496.18 (65.91%)
Total energy input[e]	MJ ha^{-1}	25027.47 (100%)

[a] Includes human labour, diesel fuel, water for irrigation, electricity.
[b] Includes seeds, chemical fertilizers, farmyard manure, biocides, machinery.
[c] Includes human labour, seeds, farmyard manure, water for irrigation.
[d] Includes diesel fuel, electricity, biocides, chemical fertilizers, machinery.
[e] Figures in parentheses indicate percentage of total energy input.

found to be 2.86. The energy ratio is often used as an index to examine the energy efficiency in crop production (Kuesters and Lammel, 1999). The energy ratio for some crops are reported as 2.8 for wheat, 4.8 for cotton, 3.8 for maize and 1.5 for sesame (Canakci et al., 2005), and 1.25 for potato (Mohammadi et al., 2008). The energy productivity of barley production was calculated as 0.19 kg MJ^{-1}. The net energy of barley production was found to be ~46,498 MJ ha^{-1}, which indicates that in this crop production, energy is gained, i.e. net energy is greater than zero. This is in agreement with results found by Nguyen and Haynes (1995), Mandal et al. (2002), Erdal et al. (2007) and Esengun et al. (2007). Khan et al. (2009) studied energy efficiency, energy productivity and specific energy for wheat, rice and barley, which amount of above indices for barley were reported as 8.21, 0.78 kg kWh^{-1} and 1.29 kWh kg^{-1}, respectively.

Also the distribution of inputs used in the production of barley according to the direct, indirect, renewable and non-renewable energy groups, are given in Table 3. It is seen that the direct and indirect energy resources are nearly equally utilized (~54% and 46%), but it is also seen that the ratios of renewable and non-renewable energy are fairly different from each other (~34% and 66%). Therefore, it is clear that non-renewable energy consumption was higher than that of renewable energy consumption in barley production, in agreement with the literature for different crops (Yilmaz et al., 2005; Erdal et al., 2007; Kizilaslan, 2009).

3.2. Econometric model estimation of barley production

In order to estimate the relationship between energy inputs and barley yield, Cobb–Douglas production function was chosen and assessed using ordinary least square estimation technique. Since the coefficient of variables in this function is in log form also represents elasticities. It is worth mentioning that elasticities and impact are the same (Mohammadi and Omid, 2010). Thus, Cobb–Douglas production function indicates a priori restriction on models of substitution among inputs.

For data used in this study, autocorrelation was tested using Durbin–Watson method (Hatrili et al., 2005). The Durbin–Watson value was found to be 2.14 for Eq. (7) which indicates that there was no autocorrelation at the 5% significance level in the estimated model. The R^2 value was determined as 0.98 for this equation, implying that around 0.98 of the variability in the energy inputs was explained by this model. Regression results for Eq. (7) were estimated and are shown in Table 4. As can be seen from Table 4, all exogenous variables had a positive impact and were found statistically significant on barley yield (expected biocides and seed energy).

Machinery had the highest impact (0.41) among other inputs and significantly contributed on the productivity at 1% level. It indicates that a 1% increase in the energy machinery input led to 0.41% increase in yield in these circumstances. The second important input was found as human labour with 0.26 elasticity followed by electricity with 0.19 elasticity. Hatrili et al. (2006) developed an econometric model for greenhouse tomato production in Antalya Province of Turkey and reported that human labour, chemical fertilizers, biocides, machinery and water energy were important inputs significantly contributed to yield.

3.3. Sensitivity of energy inputs

The sensitivity of energy inputs was analyzed using the MPP method and partial regression coefficients on output level and the results are provided in Table 4. As shown, the major MPP was drown for human labour energy (7.37), followed by machinery energy (1.66). This indicates that additional utilize of 1 MJ for each of the human labour and machinery energy would result in an increase in yield by 7.37 and 1.66 kg, respectively. Hence, exogenous parameters with large sensitivity coefficients have a strong impact on the endogenous variable. This indicates which variables should be identified and measured most carefully to assess the state of the environmental system, and which environmental factors should be managed preferentially (Drechsler, 1998). Around 74% of the total human labour energy was used on cultural practices while the remaining labour energy was used on land preparation and harvesting in barley production. About 60% of the machine energy is used for land preparation, followed by harvesting and transporting (33%), and cultural practices (7%).

Table 4
Econometric estimation and sensitivity analysis results of inputs for barley production.

Endogenous variable: yield Exogenous variables	Coefficient	t-Ratio	MPP
Model 1: $\ln\ Y_i = a_0 + \alpha_1 \ln X_1 + \alpha_2 \ln X_2 + \alpha_3 \ln X_3 + \alpha_4 \ln X_4 + \alpha_5 \ln X_5 + \alpha_6 \ln X_6 + \alpha_7 \ln X_7 + \alpha_8 \ln X_8 + e_i$			
Human labour	0.26	3.19*	7.37
Machinery	0.41	4.63*	1.66
Diesel fuel	0.12	2.15**	0.09
Total fertilizer	0.07	1.82***	0.05
Biocides	−0.04	−0.78	−1.97
Water for irrigation	0.17	2.61**	0.17
Electricity	0.19	2.83**	0.33
Seeds	−0.02	−0.52	−0.03
Durbin–Watson	2.14		
R^2	0.98		
Return to scale ($\sum_{i=1}^{n} \alpha_i$)	1.16		

* Significance at 1% level.
** Significance at 5% level.
*** Significance at 10% level.

Table 5
Econometric estimation and sensitivity analysis results of direct, indirect, renewable and non-renewable energies.

Endogenous variable: yield Exogenous variables	Coefficient	*t*-Ratio	MPP
Model 2: $\ln Y_i = \beta_1 \ln DE + \beta_2 \ln IDE + e_i$			
Direct energy	0.39	4.18*	0.13
Indirect energy	0.64	7.53*	0.28
Durbin–Watson	1.86		
R^2	0.95		
Return to scale ($\sum_{i=1}^{n} \beta_i$)	1.03		
Model 3: $\ln Y_i = \gamma_1 \ln RE + \gamma_2 \ln NRE + e_i$			
Renewable energy	0.24	3.06*	0.14
Non-renewable energy	0.81	9.28*	0.23
Durbin–Watson	1.77		
R^2	0.94		
Return to scale ($\sum_{i=1}^{n} \gamma_i$)	1.05		

* Significance at 1% level.

The MPP of biocides and seed energy were found to be −1.97 and −0.03; a negative value of MPP of inputs implies that additional units of inputs are contributing negatively to production, i.e. less production with more input. Although, the share of biocides was 0.73% of the total energy input, the use of biocides in barley production per hectare in the research area is 3.2 times higher than that of Iran's mean (Anonymous, 2007). Other important variables that influence barley yield are electricity, water for irrigation, diesel fuel, and total fertilizer energy with MPP of 0.33, 0.17, 0.09 and 0.05, respectively.

Singh et al. (2004) examined the sensitivity of energy inputs on wheat productivity for five agro-climate zones in India. They reported that MPP of chemicals in zones 1–5, were calculated to be 0.385, 2.816, −0.211, 0.610 and 0.624, respectively.

As in Table 5, the energy obtained from existing inputs was divided into two direct and indirect forms. The assessed trends of both forms of energy were positive, showing the positive impacts on the output level. Impact of indirect energy (0.64) was more than that of direct energy (0.39). This impact was significant at 1% level. The regression coefficient for renewable energy (0.24) and non-renewable energy (0.81) was significant at 1% level. It concludes that impact of non-renewable energy was higher than that of renewable energy in barley production. Mohammadi and Omid (2010) and Hatrili et al. (2005) have also reported similar results for greenhouse cucumber production in Iran and energy use in Turkish agriculture, respectively. Computed Durbin–Watson values were calculated as 1.86 and 1.77 for Eqs. (8) and (9) and the R^2 values were found to be 0.95 and 0.94, respectively.

The MPP of direct, indirect, renewable and non-renewable energy were found to be 0.13, 0.28, 0.14 and 0.23, respectively. This indicates that with an additional use of 1 MJ of each of the direct, indirect, renewable and non-renewable energy would lead to an additional increase in yield by 0.13, 0.28, 0.14 and 0.23 kg, respectively. The sum of the regression coefficients of energy inputs was calculated more than unity as 1.16, 1.03 and 1.05 for Eqs. (7)–(9), respectively. This implied that a 1% increase in the total energy inputs utilize would lead in 1.16%, 1.03% and 1.05% increase in the barley yield for these equations. Thus, there prevailed an increasing return to scale for estimated models.

Since barley production in the region is usually practiced as a single crop system and no crop rotation is planned, high use of chemical fertilizers is expected to compensate for the soil nutrients deficiency. Therefore, using proper management of energy sources and crop rotation such as growing the leguminous plants which stabilize the nitrogen in the soil can decrease its consumption. Among the reasons of high consumption of diesel fuel, the worn out machinery and the method of tillage procedures can be addressed. Using reasonable methods of tillage can minimize the machinery entrance to the farm and fuel consumption. Training the farmers to use various sources of energy in a proper way and at a suitable time, besides using tractors and machineries having the capacity and power proportional to farm size in the studied area can optimize the energy consumption further.

4. Conclusions

The aim of this study was to analyze sensitivity of a particular energy input level on barley yield in Hamedan Province, Iran. Based on the results of the investigations, the following conclusions were drawn:

1. Total fertilizer found as the most energy consuming input was followed by diesel fuel for barley production.
2. The ratio of non-renewable energy was greater than that of renewable energy consumption. Since the main non-renewable input was chemical fertilizers, management of plant nutrients by renewable resources like farmyard manure and green manures would increase rate of renewable energy.
3. The MPP estimated for human labour energy was the biggest among inputs of energy. As well, MPP of biocides energy was found negative, indicating that biocides energy consumption is high in barley production.
4. It is suggested that new policies are to be implemented to reduce the negative effects of energy inputs such as plant, soil and climate pollution. Therefore, analysis of energy consumption is an important task. These would lead to develop an established sustainability, efficient energy utilization and environment friendly agricultural production systems in the area.

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